

TrafficEye System

Traffic Monitoring & Green Light Support System



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TrafficEye System

Traffic Monitoring & Green Light Support System

Smarter Traffic. Safer Roads. Better Decisions.



Stakeholders

Traffic Police Officers	Monitor traffic, get alerts, make decisions.
Traffic Control Centres	Real-time monitoring and signal management.
Road Management Authorities	Plan and improve traffic infrastructure.
City Planners	Use data for smart city planning and development.
	Experience smoother travel, less waiting time.
	More reliable and efficient transport service.

Our Solution

TrafficEye is an AI-based system that monitors traffic density at junctions and provides smart green light timing support. It uses video analysis and real-time data to detect vehicle queues, calculate the time needed to clear the junction, and recommend the right green light duration.

- YOLOv8 vehicle detection & tracking
- Queue position & vehicle type analysis
- Clearance time calculation
- Green light recommendation
- Real-time dashboard & alerts



Why It Matters

- Reduces waiting time and congestion
- Less fuel consumption and air pollution
- Improves road safety
- More efficient traffic management

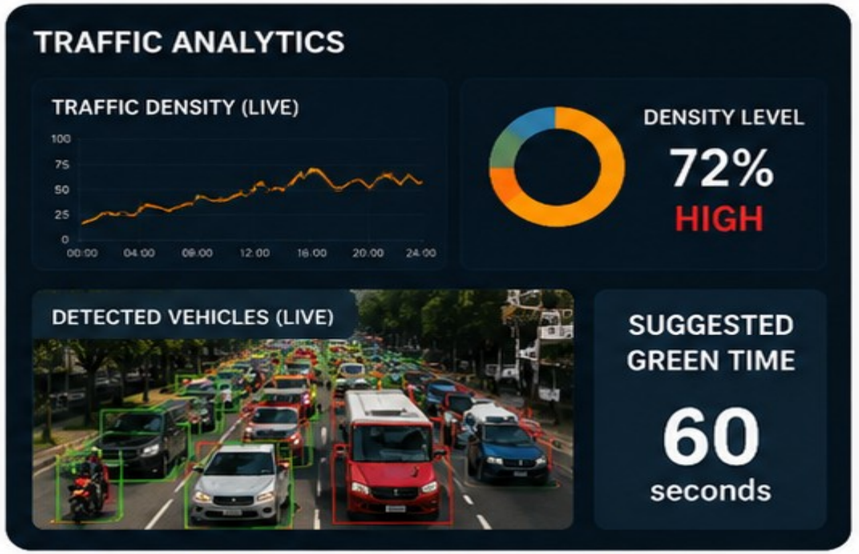
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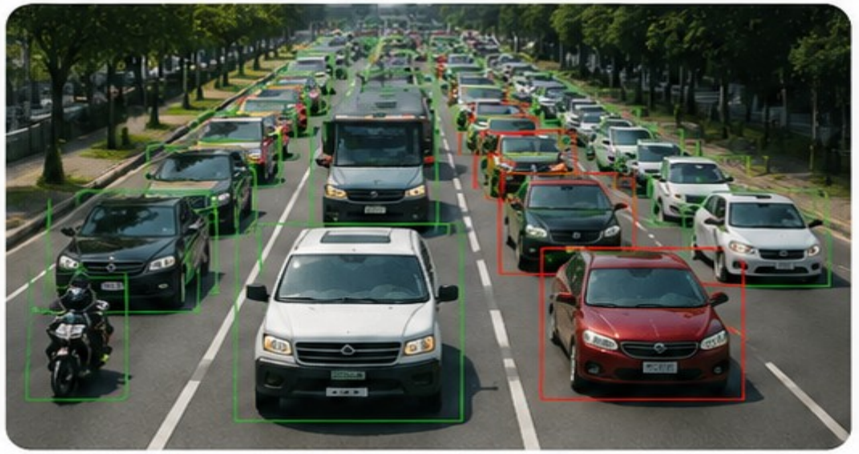
Main Objective

To design, develop, and validate a multi-modal decision support system for traffic junctions that recommends the optimal green light duration by estimating queue clearance time using AI-based vehicle detection and real-time traffic analysis, achieving accurate clearance time estimation with safety buffer integration.



Solutions Summary

- **AI-Based Vehicle Detection:** YOLOv8 model detects and classifies vehicles (Car, Bus, Truck, Motorcycle, Tuk Tuk, Ambulance) in real-time from video/CCTV.
- **Queue Analysis (FIFO Ordering):** Vehicles are ordered as P1, P2, P3... using FIFO principle to identify queue position.
- **Clearance Time Estimation:** Calculates distance, effective discharge speed, and clearance time for each vehicle.
- **Green Time Recommendation:** Computes total clearance time for the queue and adds safety buffer to recommend optimal green time.
- **Decision-Support Dashboard:** Provides real-time density level, queue length, and suggested green time for traffic officers.



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SYSTEM ARCHITECTURE

1. VIDEO INPUT & AI VEHICLE DETECTION



Traffic Video
(CCTV / IP Camera)



YOLOv8 Model
Vehicle Detection & Tracking

Detected Vehicles

- Car
- Bus
- Truck
- Motorcycle
- Tuk Tuk
- Ambulance

2. QUEUE ANALYSIS (FIFO)

Identify queue & assign positions using FIFO

- P1 (Front Vehicle)
- P2
- P3
- P4
- ⋮
- Pn (Last Vehicle)

3. CLEARANCE TIME CACLULATION

Vehicle Length (L)
Assign representative length for each vehicle type

Standstill Gap (g)
Add gap distance (1.177 m) between vehicles

Clearance Distance (D)
Calculate distance each vehicle must travel to clear junction

Effective Discharge Speed (S)
Calculate from observed distance / time

Clearance Time (T)
T = D / S (for each vehicle)



CENTRAL SERVER & DATABASE

4. WEB DASHBOARD (CONTROL CENTER)



- Real-time Monitoring
- Traffic Analytics
- Historical Reports
- Junction Management
- User & System Settings

5. GREEN LIGHT RECOMMENDATION



Final Recommended Green Time

60 seconds

Includes safety buffer for real-world uncertainties

TrafficEye System

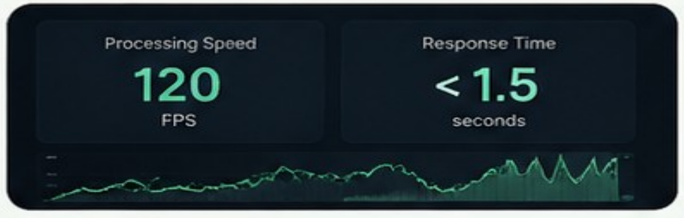
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These non-functional requirements ensure that the system is fast, reliable, easy to use, secure and transparent while supporting real-time traffic decision making.

1. Performance

Real-time video processing and analysis with fast response to support instant green light recommendations.



2. Efficiency

Optimized algorithms and resource usage to handle high traffic volume with minimal computational overhead.



3. Usability

User-friendly dashboard and clear visualizations for traffic officers to understand and make decisions quickly.



4. Reliability

Continuous operation with high accuracy and stability even under different lighting, weather and traffic conditions.



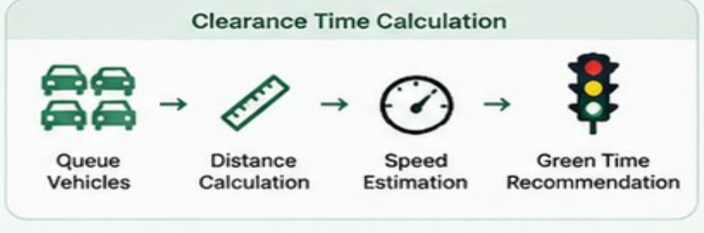
5. Security

Secure data transmission, user authentication and role-based access to protect system data and privacy.



6. Explainability

Transparent calculation logic and visual reports to explain how the recommended green time is generated.



System Architecture Layers

1 Frontend Layer (Client)



React-based web application for dashboards, live monitoring, analytics and system management.

KEY FEATURES

- Live Video Monitoring
- Traffic Analytics & Reports
- Alerts & Notifications
- System Settings

TECHNOLOGIES

- React 18, Vite 5
- Tailwind CSS v4
- Lucide-react, Material Symbols
- Leaflet, React-Leaflet (Maps)
- Context API (State Management)
- WebSocket (/ws/stream)
- ESLint 9

2 Backend Layer (Server)



FastAPI backend providing REST APIs, authentication, business logic, real-time data streaming and integrations.

KEY FEATURES

- Authentication & Authorization
- Analysis & Prediction Engine
- API Services (HTTP)
- Real-time Stream (WebSocket)

TECHNOLOGIES

- Python, FastAPI
- RESTful APIs (JSON)
- WebSocket (Live Stream)
- CORS Middleware (allow_origins=["**])
- Uvicorn ASGI Server

3 Storage Layer (Database)



Persistent storage for system data, logs, user information and analytics with offline caching support.

KEY FEATURES

- Tree Records
- Sensor Logs
- Tasks & Logs
- User & System Data

TECHNOLOGIES

MongoDB Atlas

- Scalable Cloud Database
- Secure & Reliable
- Automatic Backups

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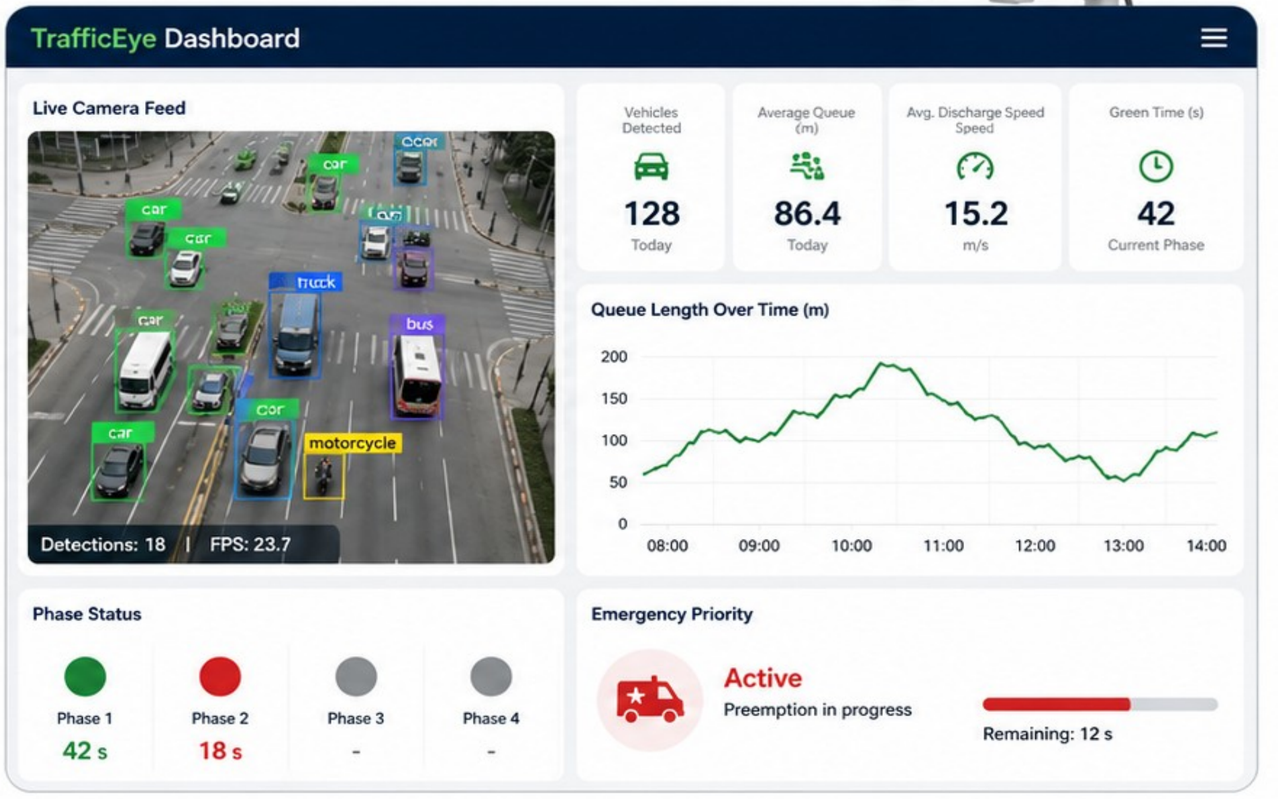
Integration & Completion Status

TrafficEye – Traffic Monitoring & Green Light Support System

Completed Components

- YOLOv8 vehicle detection
- Vehicle classification
- Queue position analysis
- Calibration pipeline
- Effective discharge speed calculation
- Final lookup table generation
- Green-time calculation module
- Emergency priority module
- Web dashboard

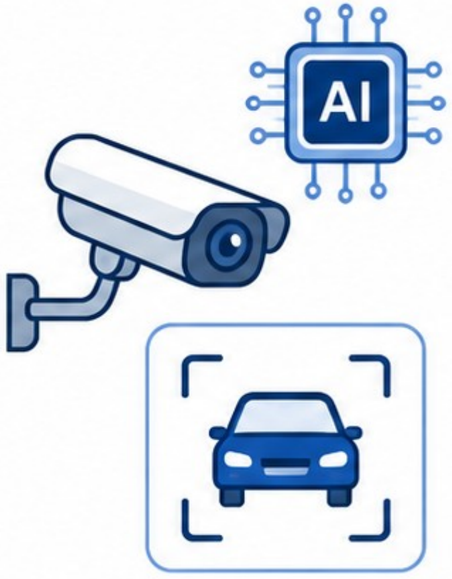




Evaluation Evidence

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1 Model Evaluation



YOLOv8 evaluated using:

- Detection accuracy
- Precision
- Recall
- mAP

2 Calibration Evaluation



System evaluated using:

- Multiple traffic clips
- Effective speed calibration

THANK YOU!